

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10: QUADRATIC EXPRESSIONS AND EQUATIONS

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This Final Explanation is about Mama Wanjiru's vegetable plot in Kericho. She has exactly 40 m of fencing wire for a rectangular plot of sukuma wiki and spinach. Her children Chebet and Kiprop found that two different widths can give the same area. Use what you learned in Lessons 1 to 8 to answer each part. Write your answers in the space under each prompt. Show your working, use units (m and m²), and check every answer against the situation. Refer to your Lesson 1 model and your final model where the questions ask you to.

Part 1: Forming the area expression

Mama Wanjiru uses all 40 m of wire for the fence of a rectangular plot. Let the width be x m.

(a) Explain why the length is $(20 - x)$ m.

(b) Write the area as an expression in x and expand it. Say why it is called a quadratic expression.

(c) Which widths x give a real plot? Explain why widths of 20 m or more do not.

(d) Use your expression to find the area when $x = 8$, $x = 10$ and $x = 12$.

(a) The fence goes all round the plot, so $2(\text{length} + \text{width}) = 40$. Therefore $\text{length} + \text{width} = 20$ m. If the width is x m, the length is $(20 - x)$ m.

(b) Area = width $\times$ length = $x(20 - x) = 20x - x^2$ (m²). It is a quadratic expression because its highest power of x is 2. The x^2 term comes from multiplying x by a length that also contains x , which makes an x by x square region in the area model.

(c) Valid widths are $0 < x < 20$. At $x = 20$ the length is $20 - 20 = 0$, so there is no plot. Above 20 the length is negative, which is impossible.

(d) $x = 8$: $8 \times 12 = 96$ m². $x = 10$: $10 \times 10 = 100$ m². $x = 12$: $12 \times 8 = 96$ m².

Part 2: Why two widths give one area

Chebet and Kiprop noticed that a plot 8 m wide and a plot 12 m wide have the same area.

(a) Write the equation you get when the area is 96 m² and

(a) $x(20 - x) = 96$ gives $20x - x^2 = 96$, so $x^2 - 20x + 96 = 0$.

(b) We need two numbers with product 96 and sum -20 . These are -8 and -12 . So $(x - 8)(x - 12) = 0$. By the zero-product property, if $p \times q = 0$ then $p = 0$ or $q = 0$, so $x - 8 = 0$ or $x - 12 = 0$, giving $x = 8$ or $x = 12$. Check: $8 \times 12 = 96$ and $12 \times 8 = 96$. Both are valid because $0 < x < 20$. Expanding $(x - 8)(x - 12)$ gives $x^2 - 20x + 96$, which also confirms the factorisation.

(c) No. When $x = 8$ the length is 12 m. When $x = 12$ the length is 8 m. Both are the same 8 m by 12 m rectangle turned round.

rearrange it so that one side is zero. (b) Factorise it and use the zero-product property to solve it. Show how you can check your answers. (c) Are the 8 m and 12 m plots two different shapes? Explain using the length and width of each plot. (d) Use the symmetry of the area table to predict one other pair of widths that give equal areas, and state that area.	Width and length add up to 20 m, so x and $20 - x$ describe the same rectangle. (d) For example $x = 6$ and $x = 14$ both give 84 m^2 ($6 \times 14 = 84$). Another correct answer is $x = 4$ and $x = 16$, which both give 64 m^2. The areas are symmetric about $x = 10$.
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Part 3: Why there is a maximum area

With 40 m of wire, the plot with a width of 10 m gives the largest area. (a) Which identities from Lessons 2 and 3 could help you show that the area $x(20 - x)$ can be written as $100 - (x - 10)^2$? Write one of them in full. (b) Show by expanding that $100 - (x - 10)^2$ is equal to $20x - x^2$. (c) Use this form to explain why no plot can have an area greater than 100 m^2, and say what shape the plot is when the area is 100 m^2.	(a) The identity $(a - b)^2 = a^2 - 2ab + b^2$ is used. The identity $a^2 - b^2 = (a - b)(a + b)$ is also part of the model. (b) $(x - 10)^2 = x^2 - 20x + 100$. So $100 - (x - 10)^2 = 100 - x^2 + 20x - 100 = 20x - x^2$. This equals $x(20 - x)$. (c) $A = 100 - (x - 10)^2$. A square $(x - 10)^2$ can never be less than 0, so we never subtract a negative amount from 100. Therefore $A \leq 100$. The area equals 100 m^2 only when $(x - 10)^2 = 0$, so $x = 10$. Then the length is also 10 m, so the plot is a 10 m by 10 m square.
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Part 4: Designing a school-garden plot

Your group has 40 m of fencing wire for a school-garden plot. (a) The target area is 84 m^2. Form the equation, solve it by factorisation, and state the dimensions of the plot. Check that it uses exactly 40 m of wire. (b) A second group wants an	(a) $x(20 - x) = 84$, so $x^2 - 20x + 84 = 0$. We need two numbers with product 84 and sum -20: -6 and -14. So $(x - 6)(x - 14) = 0$, giving $x = 6$ or $x = 14$. Both are between 0 and 20. The plot is 6 m by 14 m. Check: area = $6 \times 14 = 84 \text{ m}^2$. Wire = $2(6 + 14) = 40 \text{ m}$. (b) It is not possible. The area is $A = 100 - (x - 10)^2$, so it cannot be more than 100 m^2, and 105 m^2 is bigger than 100 m^2. There is no width that gives this area with 40 m of wire.
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area of 105 m ² . Decide whether this is possible and explain your reasoning.	
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Part 5: Using the method in other situations and answering the driving question

A mandazi tray at a Kisumu market stall holds 60 mandazi in x rows and $(x + 4)$ columns. (a) Form and solve an equation for x. State which solution you keep and why. (b) For a ball kicked at Kasarani, the height is $h = 20t - 5t^2$ metres after t seconds. The ball is at 15 m high at two times. Find both times and explain why both are valid. (c) Now answer the driving question: How can quadratic expressions and equations be used to find the dimensions of a plot when its fence length and area are both fixed, and why do two different plots sometimes have the same area? Compare what you thought in Lesson 1 with what you know now.	(a) $x(x + 4) = 60$, so $x^2 + 4x - 60 = 0$. Then $(x + 10)(x - 6) = 0$, so $x = -10$ or $x = 6$. We reject $x = -10$ because the number of rows cannot be negative. So $x = 6$: there are 6 rows of 10 mandazi ($6 \times 10 = 60$). (b) $20t - 5t^2 = 15$ gives $5t^2 - 20t + 15 = 0$, so $t^2 - 4t + 3 = 0$. Then $(t - 1)(t - 3) = 0$, so $t = 1$ s or $t = 3$ s. Check: $20 - 5 = 15$ and $60 - 45 = 15$. Both are valid because the ball is 15 m high once on the way up and once on the way down. (c) With 40 m of wire the length is $20 - x$, so the area is $x(20 - x)$. To find the widths for a target area, we set the area equal to the target, rearrange to zero, factorise and use the zero-product property. For 96 m² this gives $(x - 8)(x - 12) = 0$, so $x = 8$ or $x = 12$. Then we check the answers against $0 < x < 20$. The two plots with equal area are the same rectangle (8 m by 12 m) turned round, because width + length = 20 m, so the area is symmetric about $x = 10$. The maximum area is 100 m² at $x = 10$ because $A = 100 - (x - 10)^2$ and a square cannot be less than 0. In Lesson 1 I could only find areas by trial and error and could not explain the pairs or the maximum. Now I can solve for the widths and explain both.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Forming and expanding quadratic expressions	Correctly derives length $20 - x$ and area $x(20 - x) = 20x - x^2$, explains the x^2 term and the valid range $0 < x < 20$ with clear reasons.	Writes the correct expression and expands it, with the range given but with a limited explanation of the x^2 term or the limits.	Has errors in the length or area expression, or does not expand, or omits the valid range.
Solving by factorisation and checking	Rearranges to zero, factorises correctly, uses the zero-product property, and checks each solution against the context and by substitution or expansion.	Solves correctly but the checking against the context is brief or incomplete.	Makes factorisation or rearrangement errors, or does not use the zero-product property or check solutions.
Explaining equal areas and the	States that 8 m and 12 m give the same	Explains the equal areas or the maximum	Describes the two plots as different

maximum	8 m by 12 m rectangle turned round, explains symmetry about $x = 10$, and uses $A = 100 - (x - 10)^2$ to explain the maximum of 100 m ² .	correctly but the reasoning for one of them is incomplete.	shapes, or states the maximum without a reason.
Applying to new situations	Forms and solves the mandazi, ball and garden equations accurately, rejecting invalid roots and keeping valid ones with a reason linked to the situation.	Forms and solves most equations correctly but gives weak reasons for keeping or rejecting roots.	Cannot form the equations from the situations, or accepts invalid roots.
Communicating and linking to the driving question	Gives a clear answer to the driving question, compares the Lesson 1 and final models, and uses correct units and vocabulary throughout.	Answers the driving question with mostly correct vocabulary and units but a limited comparison to Lesson 1.	The answer to the driving question is incomplete or unclear, with frequent missing units or vocabulary errors.

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